Putnam
Single-Variable Calculus

A curated collection from the William Lowell Putnam Mathematical Competition, 1965–2025

Compiled for ISI UGB preparation

Problems on limits, continuity, differentiation, integration, series, sequences, miscellaneous

Part I. Problems from 1965–1984

1965 — The Twenty-Sixth Competition

B–4. Consider the function

$$f(x,n) = \frac{\binom{n}{0} + \binom{n}{2}x + \binom{n}{4}x^2 + \cdots}{\binom{n}{1} + \binom{n}{3}x + \binom{n}{5}x^2 + \cdots},$$

where n is a positive integer. Express $f(x, n+1)$ rationally in terms of $f(x,n)$ and x. Hence, or otherwise, evaluate $\lim_{n\to\infty} f(x,n)$ for suitable fixed values of x.

1966 — The Twenty-Seventh Competition

A–3*. Let $0 < x_1 < 1$ and $x_{n+1} = x_n(1 - x_n)$ for $n = 1, 2, 3, \ldots$. Show that $\lim_{n\to\infty} nx_n = 1$.

A–6. Justify the statement that

$$3 = \sqrt{1 + 2\sqrt{1 + 3\sqrt{1 + 4\sqrt{1 + 5\sqrt{1 + \cdots}}}}}.$$

B–3. Show that if the series $\displaystyle\sum_{n=1}^{\infty} \frac{1}{p_n}$ is convergent, where $p_1, p_2, \ldots$ are positive real numbers, then the series

$$\sum_{n=1}^{\infty} \frac{n^2}{(p_1 + p_2 + \cdots + p_n)^2}\, p_n$$

is also convergent.

1967 — The Twenty-Eighth Competition

A–1*. Let $f(x) = a_1 \sin x + a_2 \sin 2x + \cdots + a_n \sin nx$, where $a_1, \ldots, a_n$ are real and n is a positive integer. Given that $|f(x)| \le |\sin x|$ for all real x, prove that

$$|a_1 + 2a_2 + \cdots + na_n| \le 1.$$

A–4. Show that if $\lambda > \frac{1}{2}$ there does not exist a real-valued function u such that for all x in the closed interval $0 \le x \le 1$,

$$u(x) = 1 + \lambda \int_x^1 u(y)\, u(y - x)\, dy.$$

B–3*. If f and g are continuous and periodic with period 1 on the real line, then

$$\lim_{n\to\infty} \int_0^1 f(x)\, g(nx)\, dx = \Big(\int_0^1 f(x)\, dx\Big)\Big(\int_0^1 g(x)\, dx\Big).$$

1968 — The Twenty-Ninth Competition

A–1. Prove

$$\frac{22}{7} - \pi = \int_0^1 \frac{x^4(1-x)^4}{1+x^2}\, dx.$$

B–4. Show that if f is real-valued and continuous on $(-\infty, \infty)$ and the integral $\int_{-\infty}^{\infty} f(x)\, dx$ exists, then

$$\int_{-\infty}^{\infty} f\big(x - \tfrac{1}{x}\big)\, dx = \int_{-\infty}^{\infty} f(x)\, dx.$$

1969 — The Thirtieth Competition

A–4. Show that

$$\int_0^1 x^x\,dx = \sum_{n=1}^{\infty}(-1)^{n+1}n^{-n}.$$

(The integrand is taken to be 1 at $x = 0$.)

A–6*. Let a sequence $\{x_n\}$ be given, and let $y_n = x_{n-1} + 2x_n$ for $n = 2, 3, 4, \dots$. Suppose that the sequence $\{y_n\}$ converges. Prove that the sequence $\{x_n\}$ also converges.

B–3. The terms of a sequence T_n satisfy

$$T_nT_{n+1} = n \quad (n = 1, 2, 3, \dots), \qquad \lim_{n\to\infty}\frac{T_n}{T_{n+1}} = 1.$$

Show that $\pi T_1^2 = 2$.

B–5. Let $a_1 < a_2 < a_3 < \cdots$ be an increasing sequence of positive integers. Let the series $\displaystyle\sum_{n=1}^{\infty}\frac{1}{a_n}$ be convergent. For any number x, let $k(x)$ be the number of the a_n's which do not exceed x. Show that $\lim_{x\to\infty} k(x)/x = 0$.

1970 — The Thirty-First Competition

A–4*. Given a sequence $\{x_n\}$, $n = 1, 2, \dots$, such that $\lim_{n\to\infty}(x_n - x_{n-2}) = 0$, prove that

$$\lim_{n\to\infty}\frac{x_n - x_{n-1}}{n} = 0.$$

B–1*. Evaluate

$$\lim_{n\to\infty}\frac{1}{n^4}\prod_{i=1}^{2n}(n^2+i^2)^{1/n}.$$

B–2. The time-varying temperature of a certain body is given by a polynomial in the time of degree at most three. Show that the average temperature of the body between 9 a.m. and 3 p.m. can always be found by taking the average of the temperatures at two fixed times, which are independent of which polynomial occurs. Also, show that these two times are 10:16 a.m. and 1:44 p.m. to the nearest minute.

B–4. An automobile starts from rest and ends at rest, traversing a distance of one mile in one minute, along a straight road. If a governor prevents the speed of the car from exceeding ninety miles per hour, show that at some time of the traverse the acceleration or deceleration of the car was at least 6.6 ft/sec^2.

B–5. Let u_n denote the ramp function

$$u_n(x) = \begin{cases} -n & x \le -n, \\ x & -n < x \le n, \\ n & x > n, \end{cases}$$

and let F denote a real function of a real variable. Show that F is continuous if and only if $u_n \circ F$ is continuous for every n. (Here $(u_n \circ F)(x) = u_n[F(x)]$.)

1971 — The Thirty-Second Competition

A–6. Let c be a real number such that n^c is an integer for every positive integer n. Show that c is a non-negative integer.

1972 — The Thirty-Third Competition

A–3. If for a sequence $x_1, x_2, \ldots$, $\lim_{n\to\infty}(x_1 + x_2 + \cdots + x_n)/n$ exists, call this limit the C-limit of the sequence. A function $f : [0,1] \to \mathbb{R}$ is called *supercontinuous* on $[0,1]$ if the C-limit exists for the sequence $f(x_1), f(x_2), f(x_3), \ldots$ whenever the C-limit exists for $x_1, x_2, x_3, \ldots$. Find all supercontinuous functions on $[0,1]$.

A–6. Let $f(x)$ be an integrable function on $0 \le x \le 1$ and suppose

$$\int_0^1 f(x)\,dx = 0, \quad \int_0^1 xf(x)\,dx = 0, \ \ldots, \ \int_0^1 x^{n-1}f(x)\,dx = 0, \quad \int_0^1 x^n f(x)\,dx = 1.$$

Show that $|f(x)| \ge 2^n(n+1)$ on a set of positive measure.

B–2. A particle moving on a straight line starts from rest and attains a velocity v_0 after traversing a distance s_0. If the motion is such that the acceleration was never increasing, find the maximum time for the traverse.

1973 — The Thirty-Fourth Competition

A–2. Consider an infinite series whose nth term is $\pm(1/n)$, the $\pm$ signs being determined according to a pattern that repeats periodically in blocks of eight. (There are 2^8 possible patterns.) An example: $++----++$, generating the series

$$1 + \tfrac{1}{2} - \tfrac{1}{3} - \tfrac{1}{4} - \tfrac{1}{5} - \tfrac{1}{6} + \tfrac{1}{7} + \tfrac{1}{8} + \cdots.$$

(a) Show that a sufficient condition for the series to be conditionally convergent is that there be four "+" signs and four "−" signs in the block of eight.

(b) Is this sufficient condition also necessary?

A–4*. How many zeros does the function $f(x) = 2^x - 1 - x^2$ have on the real line?

B–4*. On $[0,1]$, let f have a continuous derivative satisfying $0 < f'(x) \le 1$. Also suppose $f(0) = 0$. Prove that

$$\left[\int_0^1 f(x)\,dx\right]^2 \ge \int_0^1 [f(x)]^3\,dx.$$

Show an example where equality occurs.

B–6*. On the domain $0 \le \theta \le 2\pi$, prove that $\sin^2\theta \sin(2\theta)$ takes its maximum at $\pi/3$ and $4\pi/3$ (and so its minimum at $2\pi/3$ and $5\pi/3$). Show that

$$\left|\sin^2\theta\,\sin^3(2\theta)\,\sin^3(4\theta)\cdots\sin^3(2^{n-1}\theta)\,\sin(2^n\theta)\right|$$

takes its maximum at $\theta = \pi/3$. Hence derive

$$\sin^2\theta\,\sin^2(2\theta)\,\sin^2(4\theta)\cdots\sin^2(2^n\theta) \le (3/4)^n.$$

1974 — The Thirty-Fifth Competition

A–2. A circle stands in a plane perpendicular to the ground, and a point A lies in this plane exterior to the circle and higher than its bottom. A particle starting from rest at A slides without friction down an inclined straight line until it reaches the circle. Which straight line allows descent in the shortest time? (Assume constant gravity, no relativistic effects.)

B–2*. Let $y(x)$ be a continuously differentiable real-valued function of a real variable x. Show that if $(y')^2 + y^3 \to 0$ as $x \to +\infty$, then $y(x)$ and $y'(x) \to 0$ as $x \to +\infty$.

B–5. Show that

$$1 + \frac{n}{1!} + \frac{n^2}{2!} + \cdots + \frac{n^n}{n!} > \frac{e^n}{2}$$

for every integer $n \geq 0$. You may assume Taylor's remainder formula

$$e^x - \sum_{k=0}^{n} \frac{x^k}{k!} = \frac{1}{n!} \int_0^x (x-t)^n e^t \, dt,$$

together with $n! = \int_0^\infty t^n e^{-t} \, dt$.

1975 — The Thirty-Sixth Competition

B–5. Let $f_0(x) = e^x$ and $f_{n+1}(x) = x f_n'(x)$ for $n = 0, 1, 2, \ldots$. Show that

$$\sum_{n=0}^{\infty} \frac{f_n(1)}{n!} = e^e.$$

B–6. Show that if $s_n = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}$, then

(a) $n(n+1)^{1/n} < n + s_n$ for $n > 1$, and

(b) $(n-1)n^{-1/(n-1)} < n - s_n$ for $n > 2$.

1976 — The Thirty-Seventh Competition

A–6*. Suppose $f(x)$ is a twice continuously differentiable real-valued function defined for all real numbers x and satisfying $|f(x)| \leq 1$ for all x, and $(f(0))^2 + (f'(0))^2 = 4$. Prove that there exists a real number x_0 such that $f(x_0) + f''(x_0) = 0$.

B–1*. Evaluate

$$\lim_{n\to\infty} \frac{1}{n} \sum_{k=1}^{n} \left(\left\lfloor \tfrac{2n}{k} \right\rfloor - 2 \left\lfloor \tfrac{n}{k} \right\rfloor \right).$$

Express your answer in the form $\log a - b$, with a and b positive integers. Here $\lfloor x \rfloor$ is the integer such that $\lfloor x \rfloor \leq x < \lfloor x \rfloor + 1$, and $\log x$ is the natural logarithm.

1977 — The Thirty-Eighth Competition

A–4. For $0 < x < 1$, express

$$\sum_{n=0}^{\infty} \frac{x^{2^n}}{1 - x^{2^{n+1}}}$$

as a rational function of x.

B–1*. Evaluate the infinite product $\displaystyle\prod_{n=2}^{\infty} \frac{n^3 - 1}{n^3 + 1}$.

1978 — The Thirty-Ninth Competition

A–3*. Let $p(x) = 2 + 4x + 3x^2 + 5x^3 + 3x^4 + 4x^5 + 2x^6$. For k with $0 < k < 5$, define

$$I_k = \int_0^\infty \frac{x^k}{p(x)} \, dx.$$

For which k is I_k smallest?

A–5*. Let $0 < x_i < \pi$ for $i = 1, 2, \ldots, n$ and set $x = (x_1 + x_2 + \cdots + x_n)/n$. Prove that

$$\prod_{i=1}^{n} \frac{\sin x_i}{x_i} \leq \left(\frac{\sin x}{x} \right)^n.$$

B–3. The sequence $\{Q_n(x)\}$ of polynomials is defined by $Q_1(x) = 1 + x$, $Q_2(x) = 1 + 2x$, and for $m \geq 1$,

$$Q_{2m+1}(x) = Q_{2m}(x) + (m+1)\,x\,Q_{2m-1}(x), \quad Q_{2m+2}(x) = Q_{2m+1}(x) + (m+1)\,x\,Q_{2m}(x).$$

Let x_n be the largest real solution of $Q_n(x) = 0$. Prove that $\{x_n\}$ is an increasing sequence and that $\lim_{n\to\infty} x_n = 0$.

B–5*. Find the largest A for which there exists a polynomial

$$P(x) = Ax^4 + Bx^3 + Cx^2 + Dx + E,$$

with real coefficients, satisfying $0 \leq P(x) \leq 1$ for $-1 \leq x \leq 1$.

1979 — The Fortieth Competition

A–2*. Establish necessary and sufficient conditions on the constant k for the existence of a continuous real-valued function $f(x)$ satisfying $f(f(x)) = kx^9$ for all real x.

B–1. Prove or disprove: there is at least one straight line normal to the graph of $y = \cosh x$ at a point $(a, \cosh a)$ and also normal to the graph of $y = \sinh x$ at a point $(c, \sinh c)$. (At a point on a graph, the normal line is the perpendicular to the tangent at that point.)

B–2*. Let $0 < a < b$. Evaluate

$$\lim_{t\to 0}\left\{\int_0^1 \left[bx + a(1-x)\right]^t\,dx\right\}^{1/t}.$$

The final answer should not involve any operations other than addition, subtraction, multiplication, division, and exponentiation.

1980 — The Forty-First Competition

A–1*. Let b and c be fixed real numbers and let the ten points (j, y_j), $j = 1, 2, \ldots, 10$, lie on the parabola $y = x^2 + bx + c$. For $j = 1, \ldots, 9$, let I_j be the point of intersection of the tangents to the given parabola at (j, y_j) and $(j+1, y_{j+1})$. Determine the polynomial function $y = g(x)$ of least degree whose graph passes through all nine points I_j.

A–3. Evaluate

$$\int_0^{\pi/2} \frac{dx}{1 + (\tan x)^{\sqrt{2}}}.$$

A–5*. Let $P(t)$ be a non-constant polynomial with real coefficients. Prove that the system of simultaneous equations

$$0 = \int_0^x P(t)\sin t\,dt = \int_0^x P(t)\cos t\,dt$$

has only finitely many real solutions x.

A–6*. Let C be the class of all real-valued continuously differentiable functions f on the interval $0 \leq x \leq 1$ with $f(0) = 0$ and $f(1) = 1$. Determine the largest real number u such that

$$u \leq \int_0^1 |f'(x) - f(x)|\,dx$$

for all f in C.

B–1*. For which real numbers c is $(e^x + e^{-x})/2 \leq e^{cx^2}$ for all real x?

B–3*. For which real numbers a does the sequence defined by $u_0 = a$ and $u_{n+1} = 2u_n - n^2$ have $u_n > 0$ for all $n \geq 0$?

1981 — The Forty-Second Competition

A–1. Let $E(n)$ denote the largest integer k such that 5^k is an integral divisor of the product $1^1\,2^2\,3^3\cdots n^n$. Calculate $\lim_{n\to\infty} E(n)/n^2$.

A–5*. Let $P(x)$ be a polynomial with real coefficients and form the polynomial

$$Q(x) = (x^2+1)P(x)P'(x) + x\big([P(x)]^2 + [P'(x)]^2\big).$$

Given that the equation $P(x) = 0$ has n distinct real roots exceeding 1, prove or disprove that the equation $Q(x) = 0$ has at least $2n-1$ distinct real roots.

B–1. Find

$$\lim_{n\to\infty} \frac{1}{n^5} \sum_{h=1}^{n} \sum_{k=1}^{n} \big(5h^4 - 18h^2k^2 + 5k^4\big).$$

B–2*. Determine the minimum value of

$$(r-1)^2 + \left(\frac{s}{r}-1\right)^2 + \left(\frac{t}{s}-1\right)^2 + \left(\frac{4}{t}-1\right)^2$$

for all real numbers r, s, t with $1 \le r \le s \le t \le 4$.

1982 — The Forty-Third Competition

A–1. Let V be the region in the Cartesian plane consisting of all points (x, y) satisfying the simultaneous conditions $|x| \le y \le |x| + 3$ and $y \le 4$. Find the centroid $(\bar{x}, \bar{y})$ of V.

A–3. Evaluate

$$\int_0^\infty \frac{\arctan(\pi x) - \arctan x}{x}\, dx.$$

B–3. Let p_n be the probability that $c + d$ is a perfect square when the integers c and d are selected independently at random from the set $\{1, 2, 3, \ldots, n\}$. Show that $\lim_{n\to\infty}(p_n\sqrt{n})$ exists and express this limit in the form $r(\sqrt{s} - t)$, where s and t are integers and r is a rational number.

B–5. For each $x > e^e$ define a sequence $S_x = u_0, u_1, u_2, \ldots$ recursively as follows: $u_0 = e$, while for $n \ge 0$, u_{n+1} is the logarithm of x to the base u_n. Prove that S_x converges to a number $g(x)$ and that the function g defined in this way is continuous for $x > e^e$.

1983 — The Forty-Fourth Competition

A–2. The hands of an accurate clock have lengths 3 and 4. Find the distance between the tips of the hands when that distance is increasing most rapidly.

B–5*. Let $\|u\|$ denote the distance from the real number u to the nearest integer. (For example, $\|2.8\| = .2 = \|3.2\|$.) For positive integers n, let

$$a_n = \frac{1}{n}\int_1^n \left\|\frac{n}{x}\right\| dx.$$

Determine $\lim_{n\to\infty} a_n$. You may assume the identity

$$\frac{2}{1}\cdot\frac{2}{3}\cdot\frac{4}{3}\cdot\frac{4}{5}\cdot\frac{6}{5}\cdot\frac{6}{7}\cdot\frac{8}{7}\cdot\frac{8}{9}\cdots = \frac{\pi}{2}.$$

1984 — The Forty-Fifth Competition

A–4*. A convex pentagon $P = ABCDE$, with vertices labelled consecutively, is inscribed in a circle of radius 1. Find the maximum area of P subject to the condition that the chords AC and BD be perpendicular.

B–2*. Find the minimum value of

$$(u-v)^2+\left(\sqrt{2-u^2}-\frac{9}{v}\right)^2$$

for $0<u<\sqrt{2}$ and $v>0$.

B–4. Find, with proof, all real-valued functions $y=g(x)$ defined and continuous on $[0,\infty)$, positive on $(0,\infty)$, such that for all $x>0$ the y-coordinate of the centroid of the region

$$R_x=\{(s,t)\mid 0\le s\le x,\ 0\le t\le g(s)\}$$

is the same as the average value of g on $[0,x]$.

B–6*. A sequence of convex polygons $\{P_n\}$, $n\ge 0$, is defined inductively as follows. P_0 is an equilateral triangle with sides of length 1. Once P_n has been determined, its sides are trisected; the vertices of P_{n+1} are the interior trisection points of the sides of P_n. Thus P_{n+1} is obtained by cutting corners off P_n, and P_n has $3\cdot 2^n$ sides. (P_1 is a regular hexagon with sides of length 1/3.) Express $\lim_{n\to\infty}\text{Area}(P_n)$ in the form $\sqrt{a}/b$, where a and b are positive integers.

Part II. Problems from 1985–2000

1985 — The Forty-Sixth Competition

A–3. Let d be a real number. For each integer $m\ge 0$, define a sequence $\{a_m(j)\}$, $j=0,1,2,\ldots$, by the condition

$$a_m(0)=d/2^m,\qquad a_m(j+1)=\big(a_m(j)\big)^2+2a_m(j),\quad j\ge 0.$$

Evaluate $\lim_{n\to\infty}a_n(n)$.

A–5. Let $I_m=\int_0^{2\pi}\cos(x)\cos(2x)\cdots\cos(mx)\,dx$. For which integers m, $1\le m\le 10$, is $I_m\ne 0$?

B–5. Evaluate $\displaystyle\int_0^\infty t^{-1/2}e^{-1985(t+t^{-1})}\,dt$. You may assume that $\int_{-\infty}^{\infty}e^{-x^2}\,dx=\sqrt{\pi}$.

1986 — The Forty-Seventh Competition

A–1*. Find, with explanation, the maximum value of $f(x)=x^3-3x$ on the set of all real numbers x satisfying $x^4+36\le 13x^2$.

A–3. Evaluate $\displaystyle\sum_{n=0}^{\infty}\cot^{-1}(n^2+n+1)$, where $\cot^{-1}t$ for $t\ge 0$ denotes the number θ in the interval $0<\theta\le\pi/2$ with $\cot\theta=t$.

B–1*. Inscribe a rectangle of base b and height h and an isosceles triangle of base b in a circle of radius one. For what value of h do the rectangle and triangle have the same area?

1987 — The Forty-Eighth Competition

A–3. For all real x, the real-valued function $y=f(x)$ satisfies $y''-2y'+y=2e^x$.

(a) If $f(x)>0$ for all real x, must $f'(x)>0$ for all real x?

(b) If $f'(x)>0$ for all real x, must $f(x)>0$ for all real x?

A–6. For each positive integer n, let $a(n)$ be the number of zeros in the base-3 representation of n. For which positive real numbers x does the series $\displaystyle\sum_{n=1}^{\infty}\frac{x^{a(n)}}{n^3}$ converge?

B–1. Evaluate
$$\int_2^4 \frac{\sqrt{\ln(9-x)}\,dx}{\sqrt{\ln(9-x)}+\sqrt{\ln(x+3)}}.$$

B–4*. Let $(x_1, y_1) = (0.8, 0.6)$ and let
$$x_{n+1} = x_n \cos y_n - y_n \sin y_n, \quad y_{n+1} = x_n \sin y_n + y_n \cos y_n \quad \text{for } n \geq 1.$$

For each of $\lim_{n\to\infty} x_n$ and $\lim_{n\to\infty} y_n$, prove that the limit exists and find it, or prove that the limit does not exist.

1988 — The Forty-Ninth Competition

A–1. Let R be the region of the Cartesian plane satisfying both $|x| - |y| \leq 1$ and $|y| \leq 1$. Sketch R and find its area.

A–2*. A common calculus mistake is to think the product rule gives $(fg)' = f'g'$. If $f(x) = e^{x^2}$, determine, with proof, whether there is an open interval (a, b) and a non-zero function g defined on (a, b) such that this wrong product rule holds for x in (a, b).

A–3. Determine, with proof, the set of real numbers x for which $\displaystyle\sum_{n=1}^{\infty}\left(\frac{1}{n}\csc\frac{1}{n}-1\right)^x$ converges.

A–5*. Prove that there exists a unique function f from the set $\mathbb{R}^+$ of positive real numbers to $\mathbb{R}^+$ such that $f\big(f(x)\big) = 6x - f(x)$ and $f(x) > 0$ for all $x > 0$.

B–4. Prove that if $\displaystyle\sum_{n=1}^{\infty} a_n$ is a convergent series of positive real numbers, then so is $\displaystyle\sum_{n=1}^{\infty} a_n^{n/(n+1)}$.

1989 — The Fiftieth Competition

B–3. Let f be a function on $[0, \infty)$, differentiable and satisfying $f'(x) = -3f(x) + 6f(2x)$ for $x > 0$. Assume $|f(x)| \leq e^{-\sqrt{x}}$ for $x \geq 0$, so $f(x)$ tends rapidly to 0 as x increases. For n a non-negative integer, define
$$\mu_n = \int_0^\infty x^n f(x)\,dx,$$
sometimes called the nth moment of f.

(a) Express μ_n in terms of μ_0.

(b) Prove that the sequence $\{\mu_n 3^n/n!\}$ always converges, and that the limit is 0 only if $\mu_0 = 0$.

1990 — The Fifty-First Competition

A–2. Is $\sqrt{2}$ the limit of a sequence of numbers of the form $\sqrt[3]{n} - \sqrt[3]{m}$, $n, m = 0, 1, 2, \ldots$?

B–1*. Find all real-valued continuously differentiable functions f on the real line such that for all x,
$$\big(f(x)\big)^2 = \int_0^x \Big[\big(f(t)\big)^2 + \big(f'(t)\big)^2\Big]\,dt + 1990.$$

1991 — The Fifty-Second Competition

A–3. Find all real polynomials $p(x)$ of degree $n \geq 2$ for which there exist real numbers $r_1 < r_2 < \cdots < r_n$ such that $p(r_i) = 0$ for $i = 1, \ldots, n$, and $p'\big((r_i + r_{i+1})/2\big) = 0$ for $i = 1, \ldots, n-1$.

A–5. Find the maximum value of
$$\int_0^y \sqrt{x^4 + (y - y^2)^2}\,dx$$

for $0 \le y \le 1$.

B–2*. Suppose f and g are non-constant, differentiable, real-valued functions on $\mathbb{R}$. Suppose that for each pair of real numbers x and y,

$$f(x+y) = f(x)f(y) - g(x)g(y), \qquad g(x+y) = f(x)g(y) + g(x)f(y).$$

If $f'(0) = 0$, prove that $\big(f(x)\big)^2 + \big(g(x)\big)^2 = 1$ for all x.

1992 — The Fifty-Third Competition

A–4*. Let f be an infinitely differentiable real-valued function on the real numbers. If

$$f\left(\frac{1}{n}\right) = \frac{n^2}{n^2+1}, \qquad n = 1, 2, 3, \ldots,$$

compute the values of the derivatives $f^{(k)}(0)$, $k = 1, 2, 3, \ldots$.

1993 — The Fifty-Fourth Competition

A–1*. The horizontal line $y = c$ intersects the curve $y = 2x - 3x^3$ in the first quadrant. Find c so that the areas of the two shaded regions are equal.

A–5*. Show that

$$\int_{-100}^{-10} \left(\frac{x^2 - x}{x^3 - 3x + 1}\right)^2 dx + \int_{1/101}^{1/11} \left(\frac{x^2 - x}{x^3 - 3x + 1}\right)^2 dx + \int_{101/100}^{11/10} \left(\frac{x^2 - x}{x^3 - 3x + 1}\right)^2 dx$$

is a rational number.

B–3. Two real numbers x and y are chosen at random in $(0, 1)$ with respect to the uniform distribution. What is the probability that the closest integer to x/y is even? Express the answer in the form $r + s\pi$, where r and s are rational.

B–4. The function $K(x, y)$ is positive and continuous for $0 \le x \le 1$, $0 \le y \le 1$, and the functions $f(x)$ and $g(x)$ are positive and continuous for $0 \le x \le 1$. Suppose that for all x, $0 \le x \le 1$,

$$\int_0^1 f(y)K(x,y)\,dy = g(x) \quad \text{and} \quad \int_0^1 g(y)K(x,y)\,dy = f(x).$$

Show that $f(x) = g(x)$ for $0 \le x \le 1$.

1994 — The Fifty-Fifth Competition

A–1. Suppose that a sequence $a_1, a_2, \ldots$ satisfies $0 < a_n \le a_{2n} + a_{2n+1}$ for all $n \ge 1$. Prove that the series $\displaystyle\sum_{n=1}^{\infty} a_n$ diverges.

A–2. Let A be the area of the region in the first quadrant bounded by the line $y = \frac{1}{2}x$, the x-axis, and the ellipse $\frac{1}{9}x^2 + y^2 = 1$. Find the positive number m such that A equals the area of the region in the first quadrant bounded by the line $y = mx$, the y-axis, and the same ellipse.

B–2*. For which real numbers c is there a straight line that intersects the curve $y = x^4 + 9x^3 + cx^2 + 9x + 4$ in four distinct points?

B–3*. Find the set of all real numbers k with the following property: for any positive, differentiable function f that satisfies $f'(x) > f(x)$ for all x, there exists N such that $f(x) > e^{kx}$ for all $x > N$.

1995 — The Fifty-Sixth Competition

A–2. For what pairs (a, b) of positive real numbers does the improper integral

$$\int_b^\infty \left(\sqrt{\sqrt{x+a}-\sqrt{x}}-\sqrt{\sqrt{x}-\sqrt{x-b}}\right) dx$$

converge?

B–4. Evaluate

$$\sqrt[8]{2207-\cfrac{1}{2207-\cfrac{1}{2207-\cdots}}}.$$

Express your answer in the form $(a+b\sqrt{c})/d$, with a, b, c, d integers.

1996 — The Fifty-Seventh Competition

A–6. Let $c \geq 0$ be a constant. Give a complete description, with proof, of the set of all continuous functions $f : \mathbb{R} \to \mathbb{R}$ such that $f(x) = f(x^2 + c)$ for all $x \in \mathbb{R}$.

B–2. Show that for every positive integer n,

$$\left(\frac{2n-1}{e}\right)^{\frac{2n-1}{2}} < 1 \cdot 3 \cdot 5 \cdots (2n-1) < \left(\frac{2n+1}{e}\right)^{\frac{2n+1}{2}}.$$

1997 — The Fifty-Eighth Competition

A–3. Evaluate

$$\int_0^\infty \left(x - \frac{x^3}{2} + \frac{x^5}{2\cdot 4} - \frac{x^7}{2\cdot 4\cdot 6} + \cdots\right)\left(1 + \frac{x^2}{2^2} + \frac{x^4}{2^2\cdot 4^2} + \frac{x^6}{2^2\cdot 4^2\cdot 6^2} + \cdots\right) dx.$$

B–2*. Let f be a twice-differentiable real-valued function satisfying

$$f(x) + f''(x) = -xg(x)\, f'(x),$$

where $g(x) \geq 0$ for all real x. Prove that $|f(x)|$ is bounded.

1998 — The Fifty-Ninth Competition

A–3*. Let f be a real function on the real line with continuous third derivative. Prove that there exists a point a such that

$$f(a) \cdot f'(a) \cdot f''(a) \cdot f'''(a) \geq 0.$$

B–1*. Find the minimum value of

$$\frac{(x+1/x)^6 - (x^6+1/x^6) - 2}{(x+1/x)^3 + (x^3+1/x^3)}$$

for $x > 0$.

1999 — The Sixtieth Competition

A–5*. Prove that there is a constant C such that, if $p(x)$ is a polynomial of degree 1999, then

$$|p(0)| \leq C \int_{-1}^{1} |p(x)|\, dx.$$

B–1. Right triangle ABC has right angle at C and $\angle BAC = \theta$; the point D is chosen on AB so that $|AC| = |AD| = 1$; the point E is chosen on BC so that $\angle CDE = \theta$. The perpendicular to BC at E meets AB at F. Evaluate $\lim_{\theta\to 0} |EF|$.

B–2*. Let $P(x)$ be a polynomial of degree n such that $P(x) = Q(x)P''(x)$, where $Q(x)$ is a quadratic polynomial and $P''(x)$ is the second derivative of $P(x)$. Show that if $P(x)$ has at least two distinct roots then it must have n distinct roots.

B–4*. Let f be a real function with a continuous third derivative such that $f(x)$, $f'(x)$, $f''(x)$, $f'''(x)$ are positive for all x. Suppose $f'''(x) \le f(x)$ for all x. Show that $f'(x) < 2f(x)$ for all x.

2000 — The Sixty-First Competition

A–1*. Let A be a positive real number. What are the possible values of $\displaystyle\sum_{j=0}^{\infty} x_j^2$, given that $x_0, x_1, \dots$ are positive numbers for which $\displaystyle\sum_{j=0}^{\infty} x_j = A$?

A–4. Show that the improper integral

$$\lim_{B\to\infty} \int_0^B \sin(x)\sin(x^2)\,dx$$

converges.

B–3. Let $f(t) = \sum_{j=1}^{N} a_j \sin(2\pi jt)$, where each a_j is real and $a_N \neq 0$. Let N_k denote the number of zeros (including multiplicities) of $\dfrac{d^k f}{dt^k}$ in $[0,1)$. Prove that

$$N_0 \le N_1 \le N_2 \le \cdots \quad \text{and} \quad \lim_{k\to\infty} N_k = 2N.$$

B–4*. Let $f(x)$ be a continuous function such that $f(2x^2-1) = 2xf(x)$ for all x. Show that $f(x) = 0$ for $-1 \le x \le 1$.

Part III. Problems from 2001–2025

2001 — The Sixty-Second Competition

A–6*. Can an arc of a parabola inside a circle of radius 1 have a length greater than 4?

2002 — The Sixty-Third Competition

A–1*. Let k be a fixed positive integer. The n-th derivative of $\dfrac{1}{x^k-1}$ has the form $\dfrac{P_n(x)}{(x^k-1)^{n+1}}$ where $P_n(x)$ is a polynomial. Find $P_n(1)$.

B–3. Show that, for all integers $n > 1$,

$$\frac{1}{2ne} < \frac{1}{e} - \left(1 - \frac{1}{n}\right)^n < \frac{1}{ne}.$$

2003 — The Sixty-Fourth Competition

A–3. Find the minimum value of

$$\left|\sin x + \cos x + \tan x + \cot x + \sec x + \csc x\right|$$

for real numbers x.

2004 — The Sixty-Fifth Competition

B–3. Determine all real numbers $a > 0$ for which there exists a nonnegative continuous function $f(x)$ defined on $[0, a]$ with the property that the region

$$R = \{(x, y) : 0 \le x \le a,\ 0 \le y \le f(x)\}$$

has perimeter k units and area k square units for some real number k.

B–5. Evaluate

$$\lim_{x \to 1^-} \prod_{n=0}^{\infty} \left(\frac{1 + x^{n+1}}{1 + x^n} \right)^{x^n}.$$

2005 — The Sixty-Sixth Competition

A–3. Let $p(z)$ be a polynomial of degree n all of whose zeros have absolute value 1 in the complex plane. Put $g(z) = p(z)/z^{n/2}$. Show that all zeros of $g'(z) = 0$ have absolute value 1.

A–5*. Evaluate

$$\int_0^1 \frac{\ln(x+1)}{x^2+1}\, dx.$$

2006 — The Sixty-Seventh Competition

B–5*. For each continuous function $f : [0, 1] \to \mathbb{R}$, let $I(f) = \int_0^1 x^2 f(x)\, dx$ and $J(f) = \int_0^1 x\big(f(x)\big)^2\, dx$. Find the maximum value of $I(f) - J(f)$ over all such functions f.

B–6. Let k be an integer greater than 1. Suppose $a_0 > 0$, and define

$$a_{n+1} = a_n + \frac{1}{\sqrt[k]{a_n}} \quad \text{for } n \ge 0.$$

Evaluate

$$\lim_{n \to \infty} \frac{a_n^{k+1}}{n^k}.$$

2007 — The Sixty-Eighth Competition

A–1. Find all values of α for which the curves $y = \alpha x^2 + \alpha x + \frac{1}{24}$ and $x = \alpha y^2 + \alpha y + \frac{1}{24}$ are tangent to each other.

B–2*. Suppose that $f : [0, 1] \to \mathbb{R}$ has a continuous derivative and that $\int_0^1 f(x)\, dx = 0$. Prove that for every $\alpha \in (0, 1)$,

$$\left| \int_0^\alpha f(x)\, dx \right| \le \frac{1}{8} \max_{0 \le x \le 1} |f'(x)|.$$

2008 — The Sixty-Ninth Competition

A–4. Define $f : \mathbb{R} \to \mathbb{R}$ by

$$f(x) = \begin{cases} x & \text{if } x \le e, \\ x f(\ln x) & \text{if } x > e. \end{cases}$$

Does $\displaystyle\sum_{n=1}^{\infty} \frac{1}{f(n)}$ converge?

B–2. Let $F_0(x) = \ln x$. For $n \geq 0$ and $x > 0$, let $F_{n+1}(x) = \int_0^x F_n(t)\,dt$. Evaluate

$$\lim_{n\to\infty} \frac{n!\,F_n(1)}{\ln n}.$$

2009 — The Seventieth Competition

B–2. A game involves jumping to the right on the real number line. If a and b are real numbers and $b > a$, the cost of jumping from a to b is $b^3 - ab^2$. For what real numbers c can one travel from 0 to 1 in a finite number of jumps with total cost exactly c?

2010 — The Seventy-First Competition

A–6. Let $f : [0, \infty) \to \mathbb{R}$ be a strictly decreasing continuous function such that $\lim_{x\to\infty} f(x) = 0$. Prove that

$$\int_0^\infty \frac{f(x) - f(x+1)}{f(x)}\,dx$$

diverges.

2011 — The Seventy-Second Competition

A–3. Find a real number c and a positive number L for which

$$\lim_{r\to\infty} \frac{r^c \int_0^{\pi/2} x^r \sin x\,dx}{\int_0^{\pi/2} x^r \cos x\,dx} = L.$$

2012 — The Seventy-Third Competition

A–3*. Let $f : [-1, 1] \to \mathbb{R}$ be a continuous function such that

(i) $f(x) = \dfrac{2 - x^2}{2} f\left(\dfrac{x^2}{2 - x^2}\right)$ for every x in $[-1, 1]$,

(ii) $f(0) = 1$,

(iii) $\displaystyle\lim_{x\to 1^-} \frac{f(x)}{\sqrt{1-x}}$ exists and is finite.

Prove that f is unique, and express $f(x)$ in closed form.

B–4*. Suppose that $a_0 = 1$ and that $a_{n+1} = a_n + e^{-a_n}$ for $n = 0, 1, 2, \ldots$. Does $a_n - \log n$ have a finite limit as $n \to \infty$?

2013 — The Seventy-Fourth Competition

A–3. Suppose that the real numbers $a_0, a_1, \ldots, a_n$ and x, with $0 < x < 1$, satisfy

$$\frac{a_0}{1-x} + \frac{a_1}{1-x^2} + \cdots + \frac{a_n}{1-x^{n+1}} = 0.$$

Prove that there exists a real number y with $0 < y < 1$ such that

$$a_0 + a_1 y + \cdots + a_n y^n = 0.$$

B–2. Let $\mathcal{C} = \bigcup_{N=1}^\infty \mathcal{C}_N$, where $\mathcal{C}_N$ denotes the set of those "cosine polynomials" of the form

$$f(x) = 1 + \sum_{n=1}^N a_n \cos(2\pi n x)$$

for which:

(i) $f(x) \geq 0$ for all real x, and

(ii) $a_n = 0$ whenever n is a multiple of 3.

Determine the maximum value of $f(0)$ as f ranges through $\mathcal{C}$, and prove that this maximum is attained.

B–4. For any continuous real-valued function f defined on the interval $[0,1]$, let

$$\mu(f) = \int_0^1 f(x)\,dx, \quad \operatorname{Var}(f) = \int_0^1 (f(x) - \mu(f))^2\,dx, \quad M(f) = \max_{0 \leq x \leq 1} |f(x)|.$$

Show that if f and g are continuous real-valued functions defined on $[0,1]$, then

$$\operatorname{Var}(fg) \leq 2\operatorname{Var}(f)\,M(g)^2 + 2\operatorname{Var}(g)\,M(f)^2.$$

2014 — The Seventy-Fifth Competition

A–1. Prove that every nonzero coefficient of the Taylor series of

$$(1 - x + x^2)\,e^x$$

about $x = 0$ is a rational number whose numerator (in lowest terms) is either 1 or a prime number.

A–3. Let $a_0 = 5/2$ and $a_k = a_{k-1}^2 - 2$ for $k \geq 1$. Compute

$$\prod_{k=0}^{\infty}\left(1 - \frac{1}{a_k}\right)$$

in closed form.

A–5*. Let

$$P_n(x) = 1 + 2x + 3x^2 + \cdots + nx^{n-1}.$$

Prove that the polynomials $P_j(x)$ and $P_k(x)$ are relatively prime for all positive integers j and k with $j \neq k$.

B–2*. Suppose that f is a function on the interval $[1,3]$ such that $-1 \leq f(x) \leq 1$ for all x and $\int_1^3 f(x)\,dx = 0$. How large can $\displaystyle\int_1^3 \frac{f(x)}{x}\,dx$ be?

B–4. Show that for each positive integer n, all the roots of the polynomial

$$\sum_{k=0}^{n} 2^{k(n-k)} x^k$$

are real numbers.

2015 — The Seventy-Sixth Competition

A–1*. Let A and B be points on the same branch of the hyperbola $xy = 1$. Suppose that P is a point lying between A and B on this hyperbola, such that the area of the triangle APB is as large as possible. Show that the region bounded by the hyperbola and the chord AP has the same area as the region bounded by the hyperbola and the chord PB.

B–1*. Let f be a three times differentiable function (defined on $\mathbb{R}$ and real-valued) such that f has at least five distinct real zeros. Prove that $f + 6f' + 12f'' + 8f'''$ has at least two distinct real zeros.

2016 — The Seventy-Seventh Competition

A–1. Find the smallest positive integer j such that for every polynomial $p(x)$ with integer coefficients and for every integer k, the integer

$$p^{(j)}(k) = \left. \frac{d^j}{dx^j} p(x) \right|_{x=k}$$

(the j-th derivative of $p(x)$ at k) is divisible by 2016.

A–3. Suppose that f is a function from $\mathbb{R}$ to $\mathbb{R}$ such that

$$f(x) + f\left(1 - \frac{1}{x}\right) = \arctan x$$

for all real $x \neq 0$. (As usual, $y = \arctan x$ means $-\pi/2 < y < \pi/2$ and $\tan y = x$.) Find

$$\int_0^1 f(x)\, dx.$$

A–6*. Find the smallest constant C such that for every real polynomial $P(x)$ of degree 3 that has a root in the interval $[0, 1]$,

$$\int_0^1 |P(x)|\, dx \leq C \max_{x \in [0,1]} |P(x)|.$$

B–1. Let $x_0, x_1, x_2, \ldots$ be the sequence such that $x_0 = 1$ and for $n \geq 0$,

$$x_{n+1} = \ln(e^{x_n} - x_n)$$

(as usual, the function ln is the natural logarithm). Show that the infinite series

$$x_0 + x_1 + x_2 + \cdots$$

converges and find its sum.

B–6. Evaluate

$$\sum_{k=1}^{\infty} \frac{(-1)^{k-1}}{k} \sum_{n=0}^{\infty} \frac{1}{k \cdot 2^n + 1}.$$

2017 — The Seventy-Eighth Competition

A–3*. Let a and b be real numbers with $a < b$, and let f and g be continuous functions from $[a, b]$ to $(0, \infty)$ such that $\int_a^b f(x)\, dx = \int_a^b g(x)\, dx$ but $f \neq g$. For every positive integer n, define

$$I_n = \int_a^b \frac{(f(x))^{n+1}}{(g(x))^n}\, dx.$$

Show that $I_1, I_2, I_3, \ldots$ is an increasing sequence with $\lim_{n\to\infty} I_n = \infty$.

B–4. Evaluate the sum

$$\begin{aligned} &\sum_{k=0}^{\infty} \left(3 \cdot \frac{\ln(4k+2)}{4k+2} - \frac{\ln(4k+3)}{4k+3} - \frac{\ln(4k+4)}{4k+4} - \frac{\ln(4k+5)}{4k+5} \right) \\ &= 3 \cdot \frac{\ln 2}{2} - \frac{\ln 3}{3} - \frac{\ln 4}{4} - \frac{\ln 5}{5} + 3 \cdot \frac{\ln 6}{6} - \frac{\ln 7}{7} - \frac{\ln 8}{8} - \frac{\ln 9}{9} + \cdots . \end{aligned}$$

2018 — The Seventy-Ninth Competition

A–3*. Determine the greatest possible value of $\sum_{i=1}^{10} \cos(3x_i)$ for real numbers $x_1, x_2, \ldots, x_{10}$ satisfying $\sum_{i=1}^{10} \cos(x_i) = 0$.

A–5*. Let $f : \mathbb{R} \to \mathbb{R}$ be an infinitely differentiable function satisfying $f(0) = 0$, $f(1) = 1$, and $f(x) \geq 0$ for all $x \in \mathbb{R}$. Show that there exist a positive integer n and a real number x such that $f^{(n)}(x) < 0$.

B–2. Let n be a positive integer, and let

$$f_n(z) = n + (n-1)z + (n-2)z^2 + \cdots + z^{n-1}.$$

Prove that f_n has no roots in the closed unit disk $\{z \in \mathbb{C} : |z| \leq 1\}$.

2019 — The Eightieth Competition

A–6. Let g be a real-valued function that is continuous on the closed interval $[0,1]$ and twice differentiable on the open interval $(0,1)$. Suppose that for some real number $r > 1$,

$$\lim_{x \to 0^+} \frac{g(x)}{x^r} = 0.$$

Prove that either

$$\lim_{x \to 0^+} g'(x) = 0 \quad \text{or} \quad \limsup_{x \to 0^+} x^r |g''(x)| = \infty.$$

B–2. For all $n \geq 1$, let

$$a_n = \sum_{k=1}^{n-1} \frac{\sin\left(\frac{(2k-1)\pi}{2n}\right)}{\cos^2\left(\frac{(k-1)\pi}{2n}\right)\cos^2\left(\frac{k\pi}{2n}\right)}.$$

Determine $\lim_{n \to \infty} \frac{a_n}{n^3}$.

2021 (February) — The Eighty-First Competition

A–3. Let $a_0 = \pi/2$, and let $a_n = \sin(a_{n-1})$ for $n \geq 1$. Determine whether

$$\sum_{n=1}^{\infty} a_n^2$$

converges.

A–6. For a positive integer N, let f_N be the function defined by

$$f_N(x) = \sum_{n=0}^{N} \frac{N + \frac{1}{2} - n}{(N+1)(2n+1)} \sin\big((2n+1)x\big).$$

Determine the smallest constant M such that $f_N(x) \leq M$ for all N and all real x.

2021 (December) — The Eighty-Second Competition

A–2*. For every positive real number x, let

$$g(x) = \lim_{r \to 0} \big((x+1)^{r+1} - x^{r+1}\big)^{1/r}.$$

Find $\lim_{x\to\infty} \frac{g(x)}{x}$.

2022 — The Eighty-Third Competition

B–1. Suppose that $P(x) = a_1x + a_2x^2 + \cdots + a_nx^n$ is a polynomial with integer coefficients, with a_1 odd. Suppose that $e^{P(x)} = b_0 + b_1x + b_2x^2 + \cdots$ for all x. Prove that b_k is nonzero for all $k \geq 0$.

2023 — The Eighty-Fourth Competition

A–1*. For a positive integer n, let $f_n(x) = \cos(x)\cos(2x)\cos(3x)\cdots\cos(nx)$. Find the smallest n such that $|f_n''(0)| > 2023$.

A–3*. Determine the smallest positive real number r such that there exist differentiable functions $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$ satisfying

(a) $f(0) > 0$,

(b) $g(0) = 0$,

(c) $|f'(x)| \leq |g(x)|$ for all x,

(d) $|g'(x)| \leq |f(x)|$ for all x, and

(e) $f(r) = 0$.

B–4. For a nonnegative integer n and a strictly increasing sequence of real numbers $t_0, t_1, \ldots, t_n$, let $f(t)$ be the corresponding real-valued function defined for $t \geq t_0$ by the following properties:

(a) $f(t)$ is continuous for $t \geq t_0$, and is twice differentiable for all $t > t_0$ other than $t_1, \ldots, t_n$;

(b) $f(t_0) = 1/2$;

(c) $\lim_{t\to t_k^+} f'(t) = 0$ for $0 \leq k \leq n$;

(d) For $0 \leq k \leq n-1$, we have $f''(t) = k+1$ when $t_k < t < t_{k+1}$, and $f''(t) = n+1$ when $t > t_n$.

Considering all choices of n and $t_0, t_1, \ldots, t_n$ such that $t_k \geq t_{k-1} + 1$ for $1 \leq k \leq n$, what is the least possible value of T for which $f(t_0 + T) = 2023$?

2024 — The Eighty-Fifth Competition

B–3. Let r_n be the nth smallest positive solution to $\tan x = x$, where the argument of tangent is in radians. Prove that

$$0 < r_{n+1} - r_n - \pi < \frac{1}{(n^2+n)\pi} \quad \text{for } n \geq 1.$$

B–6. For a real number a, let

$$F_a(x) = \sum_{n\geq 1} n^a e^{2n} x^{n^2} \quad \text{for } 0 \leq x < 1.$$

Find a real number c such that

$$\lim_{x\to 1^-} F_a(x)\, e^{-1/(1-x)} = 0 \ \text{ for all } a < c, \quad \text{and} \quad \lim_{x\to 1^-} F_a(x)\, e^{-1/(1-x)} = \infty \ \text{ for all } a > c.$$

2025 — The Eighty-Sixth Competition

A–2*. Find the largest real number a and the smallest real number b such that

$$ax(\pi - x) \leq \sin x \leq bx(\pi - x)$$

for all x in the interval $[0, \pi]$.

B–2. Let $f : [0, 1] \to [0, \infty)$ be strictly increasing and continuous. Let R be the region bounded by $x = 0$, $x = 1$, $y = 0$, and $y = f(x)$. Let x_1 be the x-coordinate of the centroid of R. Let x_2 be the x-coordinate of the centroid of the solid generated by rotating R around the x-axis. Prove that $x_1 < x_2$.